

RATIO: A Benchmark for Retrieval Across Typed Ideation Operations in Scientific Literature

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Abstract

Retrieved scientific literature can serve as inspiration for both human and AI scientists. Inspiration can take different forms: prior work may directly suggest how to address a problem, or surface directions at different levels of abstraction—zooming out to a more general view or zooming in to a concrete realization. We introduce RATIO (Retrieval Across Typed Ideation Operations), a large-scale benchmark in which relevance is defined by three operations which we name *ideation moves*: ADDRESS retrieves potential approaches for stated problems, BROADEN retrieves more general formulations, and SPECIFY retrieves concrete instantiations. RATIO is constructed from millions of full-text scientific papers across CS literature via a general recipe that extends discourse-marker distant supervision—previously used only for classification—to corpus-scale retrieval, combined with extensive LLM and human vetting. Experiments show that operation-specific fine-tuning substantially boosts retrievers but leaves much room for further improvements. RATIO provides a scalable training and evaluation framework for retrieval components that support literature-grounded ideation, opening up new research avenues in scientific inspiration retrieval.

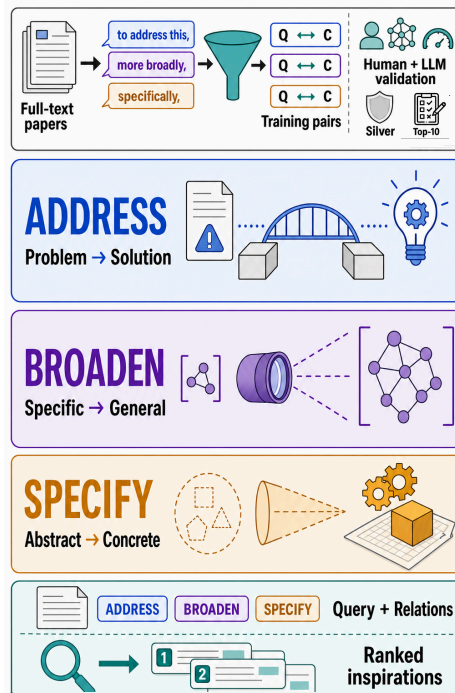


Figure 1: We construct RATIO from full-text scientific papers using discourse-marker supervision and human–LLM validation to support retrieval across three ideation moves: ADDRESS, BROADEN, SPECIFY.

1 Introduction

Engaging with prior work is central to how researchers develop new ideas: the literature can spark analogical transfer, conceptual recombination, and problem reframing (Sweed et al., 2025; Sternlicht and Hope, 2026). Consider for example a researcher grappling with reward hacking in RL-trained LLMs. Three different inspirations may be helpful: one proposing a method (e.g., reward-model ensembles), one zooming out to a more general principle (e.g., Goodhart’s law), and one zooming in on a concrete instantiation of that principle in a new setting (e.g., recommender systems gaming

engagement metrics) to provide an inspiration from another problem. Each one enables a distinct move through the ideation space—addressing a problem, abstracting it, or instantiating it.

This distinction matters for systems that assist scientific research with literature-grounded hypothesis generation (Wang et al., 2024; Garikaparathi et al., 2025; Radensky et al., 2026; Vasu et al., 2025; Liu et al., 2026; Gu et al., 2026). In this setting, retrieval determines which prior mechanisms, abstractions, and problem formulations are available as inspirations to move the ideation process forward. For both human researchers and AI agents, retrieval helps define the reachable ideation space; showing inspirational stimuli for problem-solving

in the form of potentially related mechanisms at different levels of abstraction is known to boost ideation (Hope et al., 2022; Sweed et al., 2025).

Most scholarly retrieval benchmarks, however, evaluate topical relevance or whether a document answers a search query (Ajith et al., 2024). These criteria are necessary for literature search but do not distinguish between passages that play different ideation roles. We introduce **RATIO** (Retrieval Across Typed Ideation Operations), a benchmark for scientific inspiration retrieval (Figure 1). Grounded in cognitive theories of ideation, we view external inspirations as possible moves through an ideation space (Dorst and Cross, 2001; Goldschmidt, 2014). Under this view, a retrieved statement is useful when it suggests a particular next move. **RATIO** organizes retrieval into three moves or operation-specific tasks that capture complementary forms of inspiration. **ADDRESS** retrieves an approach or insight that responds to a problem in the query. **BROADEN** retrieves a formulation at a broader scope or greater generality. **SPECIFY** retrieves a concrete instance or narrower formulation. Within each task, relevance is determined by latent relational patterns with often limited surface expression, making the desired query–candidate relation difficult for retrieval systems to recover (Tchuindjo et al., 2026).

Constructing such a benchmark at the scale required for retriever training and evaluation is challenging. We propose a scalable distant supervision approach that uses explicit high-precision discourse markers in full-text papers. Phrases such as *to address this problem*, *more generally*, and *as an example* identify adjacent sentences that authors themselves present as problem-addressing, broader, or more specific. Unlike prior discourse-marker distant supervision, which is confined to sentence-pair classification (Sadat and Caragea, 2022; Sileo et al., 2019), we extend markers to define relation-conditioned ideation retrieval over a corpus of millions of candidates (§2).

RATIO is constructed through a multi-stage supervision and validation pipeline. We first build move-specific lexicons of discourse cues for **ADDRESS**, **BROADEN**, and **SPECIFY**, combining manual corpus analysis, rule-based expansion, and generation by multiple LLMs. Candidate cues are filtered through expert review and LLM validation before being used to mine sentence pairs from full-text papers. We further create a higher-precision silver evaluation set, using human vetting of dis-

course markers and human-calibrated LLM judgments. Finally, we also validate top-10 retrieved candidates to account for false negatives.

Experiments with retrievers show that the proposed operations are not captured reliably by generic models, and operation fine-tuning yields substantial improvements for all three operations, while absolute performance remains limited.

Our contributions:

- We define *scientific ideation-move inspiration retrieval* tasks, where relevance is specified by an ideation operation: addressing a problem, broadening a formulation, or providing a concrete instantiation.
- We introduce a general methodology for constructing relation-conditioned retrieval benchmarks from raw corpora via discourse-marker supervision—including a multi-stage construction pipeline and human-calibrated LLM validation—extending marker-based distant supervision from its traditional classification setting to retrieval.
- We instantiate this methodology as **RATIO**, a large-scale full-text benchmark for the three ideation operations, with temporally held-out, high-quality evaluation sets with human vetting. We benchmark retrievers, showing that relation-specific fine-tuning substantially improves results and finds valid cross-paper inspiration candidates while leaving a considerable performance gap.

2 Related Work

Literature-grounded scientific ideation A growing body of work uses scientific literature to support hypothesis generation and research planning (Wang et al., 2024; Hu et al., 2025; Vasu et al., 2025; Liu et al., 2026; Gu et al., 2026). These tasks evaluate generation or reasoning conditioned on scientific evidence; **RATIO** focuses on the retrieval capability that determines which inspirations are made available to such systems.

Hope et al. (2022) and Sweed et al. (2025) represented problems, mechanisms, and abstractions through functional concept graphs, enabling cross-domain navigation from problems to potentially relevant mechanisms. **CHIMERA** mined and retrieved naturally occurring recombinations of scientific ideas (Sternlicht and Hope, 2026). **RATIO** shares the overarching motivation that scientific relevance extends beyond topical similarity, and focuses on introducing a novel automated benchmark for sentence-level retrieval under three

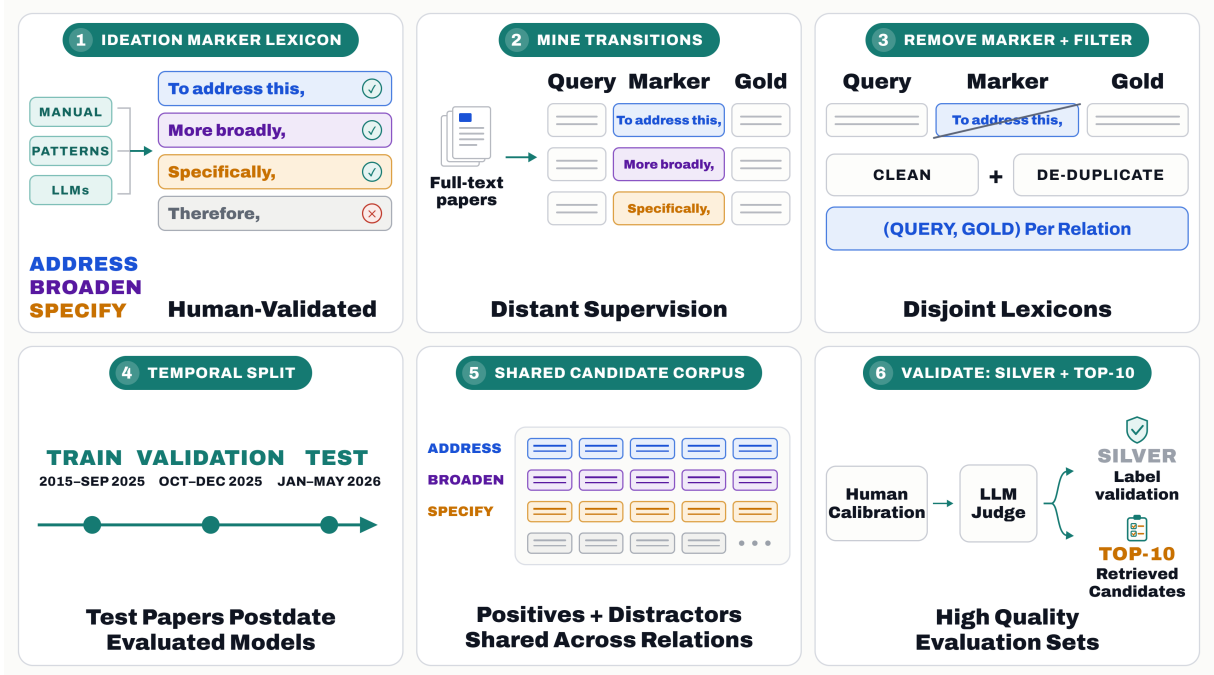


Figure 2: Construction of RATIO. Validated discourse markers identify ADDRESS, BROADEN, and SPECIFY transitions in full-text papers. The marker is removed from the candidate text, publication dates determine the temporal partitions, and all operations retrieve from a candidate corpus shared across relations.

explicit ideation-move operations, for addressing problems and zooming in and out (abstraction). In recent work, Garikaparathi et al. (2025) formulated methodology-inspiration retrieval at the paper level primarily over abstracts, using research proposals extracted from abstracts as queries and citation-derived methodological lineage as supervision. RATIO extends this direction to full-text, sentence-level retrieval and decomposes inspirational relevance into three typed operations: ADDRESS, BROADEN, and SPECIFY. Like MIR, our focus is on retrieval as the principal capability under evaluation rather than measuring downstream scientific ideation.

Scientific retrieval Scholarly retrieval benchmarks such as LitSearch evaluate the retrieval of papers satisfying literature-search questions (Ajith et al., 2024), and thus primarily characterize topical or informational relevance. RATIO instead conditions relevance on the ideation function the candidate serves: addressing a stated problem, broadening its formulation, or supplying a concrete instantiation. We also contribute an automated construction approach in which relevance labels are mined from discourse signals rather than annotations, citations, or search logs.

Scientific sentence relations SciNLI introduces natural-language inference relations for scientific

text, and MSciNLI extends scientific NLI across multiple domains (Sadat and Caragea, 2022, 2024). These resources only *classify* given sentence pairs, and only use *entailment*-oriented labels: they do not evaluate the more challenging and realistic setting of retrieval from a large corpus, and their relations do not represent our ideation motivation: problem-addressing utility or movement in opposite directions along an abstraction hierarchy. In addition, the set of discourse markers used in these resources is highly limited in size and scope, only a tiny fraction of the markers in our work.

Discourse markers as distant supervision Beyond scientific NLI, discourse markers have long served as distant supervision in the general domain. For example, Sileo et al. (2019) predict markers or marker categories to learn sentence representations, using 174 automatically discovered markers mined from web text. In this line of work, the marker is the prediction target in a sentence-pair task. RATIO instead uses markers to define relevance: the marker is stripped from the input and determines which relation a candidate must instantiate with respect to a query, evaluated by ranking against a large-scale corpus of sentences.

3 Problem Definition

We formulate the task of scientific ideation-move inspiration retrieval. Given a scientific statement and a desired move through the ideation space, the task is to retrieve a candidate inspiration statement corresponding to that move.

Formally, a *query* q may be a sentence describing a scientific research problem, claim, observation, or methodological objective; a *relation* $r \in \mathcal{R} = \{\text{ADDRESS}, \text{BROADEN}, \text{SPECIFY}\}$ is the discourse relation that must hold between two sentences; and a *scientific literature corpus* $\mathcal{C} = \{c_1, \dots, c_N\}$ is the set of all candidate sentences from scientific papers. Given q and r , the task is to produce a ranking under the relevance criterion associated with r

$$\hat{\mathcal{C}}_r^k(q) = \langle \hat{c}_1, \dots, \hat{c}_k \rangle \quad (1)$$

of the top- k sentences in $\mathcal{C}$ by how well each candidate c instantiates r with respect to q .

We define the relations as follows, for a query q and a candidate $c \in \mathcal{C}$:

- **ADDRESS:** q articulates a problem, and c an approach or insight that can help address it.
- **BROADEN:** c reformulates q at a broader scope or greater generality, such that q can be viewed as a particular instance or manifestation of c .
- **SPECIFY:** c instantiates q through a concrete case, mechanism, or example.

BROADEN and SPECIFY represent opposite movements along an abstraction hierarchy, while ADDRESS represents a functional response to a stated research need. In our experiments, we train a separate retriever s_r for each operation, while our task formulation and data collection readily support training a relation-conditioned scoring function $s(q, c \mid r)$; see Section 5 for a discussion on the rationale behind this. Throughout, we use “inspiration” operationally to denote a statement that validly instantiates the requested ideation move. As in recent work (Garikaparthi et al., 2025), RATIO focuses its evaluation on this retrieval capability, not the downstream effect of retrieved statements on idea generation or their novelty to a particular researcher.

4 Benchmark Construction

Authors routinely signal how a sentence relates to its predecessor with discourse markers (e.g., “To mitigate these risks,” “Even more specifically,”

“From a broader perspective,”). We exploit this to harvest tuples (q, ℓ, g) : in “Our model is prone to overfitting. To address this issue, we apply dropout.”, the first sentence is the query q , “To address this issue” is the marker ℓ , and the remaining proposition “we apply dropout” is the gold sentence g . For each relation r we construct a lexicon L_r , the set of validated markers whose presence signals $(q, g) \in r$. Lexicons are pairwise disjoint, so every marker determines a unique relation (§4.1). These markers provide high-precision, author-generated distant supervision for the target relations. For example, “To address this issue,” explicitly indicates that the author presents the following statement as a response to the preceding problem.

We segment each paper P into sentences and scan consecutive pairs (s_i, s_{i+1}) . Whenever the second sentence decomposes as $s_{i+1} = \ell \oplus g$ with $\ell \in L_r$ (where $\oplus$ denotes concatenation), we extract the tuple (s_i, ℓ, g) . Writing $\mathcal{E}_r(P)$ for the set of all tuples mined from paper P under relation r :

$$\mathcal{E}_r(P) = \{(s_i, \ell, g) \mid s_{i+1} = \ell \oplus g, \ell \in L_r\}. \quad (2)$$

The marker is stripped from the gold sentence and kept only as construction metadata.

We work with a corpus of full-text papers (Kinney et al., 2023)¹. Each paper is segmented into sentences, cleaned and normalized, and split into consecutive pairs (s_i, s_{i+1}) that will later become candidates for (q, g) (Appendix B).

4.1 Ideation Move Marker Lexicons

The ideation move discourse marker lexicon L is the union of the per-relation lexicons,

$$L = \bigcup_{r \in \mathcal{R}} L_r = L_{\text{ADDRESS}} \cup L_{\text{BROADEN}} \cup L_{\text{SPECIFY}}. \quad (3)$$

Each L_r is built by an iterative, multi-stage process that we describe next (see Appendix A, B for full details and examples).

Manual curation In scientific writing, when (implicitly) referring to our ideation moves, authors often use linking phrases that end with a comma at the start of a sentence. We take each sentence’s leading pre-comma span (≤ 7 words), since sentence-initial markers are typically comma-delimited, isolating them from substantive content. Spans are

¹Snapshot of May 5th, 2026.

Relation	Train		Validation		Test	
	Queries	Candidates	Queries	Candidates	Queries	Candidates
SPECIFY	2,605,515		79,583		94,079	
ADDRESS	195,605	13,787,834	13,082	361,172	14,020	404,371
BROADEN	13,687		495		1,410	

Table 1: **Temporal split with a shared candidate pool across relations.** Each query is paired with a single gold candidate; the candidate corpus of each split is shared by all relations. Train uses papers from 2015 to Sept. 30th, 2025; validation uses Oct. 1st to Dec. 31st, 2025; test uses Jan. 1st to May 5th, 2026 (Appendix B).

counted across all sentences and the 400 most frequent are manually inspected as marker candidates.

For each candidate, two NLP experts inspected 15 randomly sampled sentence pairs containing it. A marker was retained if in every inspected pair it clearly signaled the target operation and referred to the preceding sentence. For example, “*To mitigate this limitation,*” is added to ADDRESS since it satisfies both conditions: it commits the following sentence to a remedy, and its demonstrative explicitly resolves to the preceding sentence. In contrast, “*Therefore,*”, used in prior work (Sadat and Caragea, 2022) for creating NLI pairs, does not signal any of our three target ideation moves.

“*In general,*” does signal generalization, but is non-anaphoric: it does not explicitly refer to any particular preceding sentence. In total, 84 markers passed both experts for ADDRESS, 13 for BROADEN, and 72 for SPECIFY (see Appendix A).

We also manually selected a small set of discourse markers, for which we automatically added their extensions as markers. For example, we selected the marker “*To address*”, for which all extensions (of at most five consecutive words up to a comma) were collected. That is, “*To address this potential bias,*”, “*To address this limitation,*” etc. were all added as markers.

LLM expansion We expanded the manually curated markers with several SOTA LLMs, using more than one model for diversity. An LLM labels all 4,256 generated candidates as *strong-true*, *weak-true*, or *false* (3,779, 270, and 207 respectively), according to whether it estimated the marker to hold in almost all contexts, only in some, or in none. Only *strong-true* markers were kept, contributing 3,779 of the 4,252 markers in the final lexicon (89%). An NLP expert then further screened a random sample of 1,500 markers, confirming that each signals the target operation and plausibly refers to the preceding sentence.

Rule-Based Pattern Expansion Similarly to Roller et al. (2018), who apply Hearst-style tem-

Type	ADDRESS		BROADEN		SPECIFY		Total	
	Cand.	Data	Cand.	Data	Cand.	Data	Cand.	Data
Manual	84	49	13	7	72	52	169	108
Extendable connective	5	4	0	0	2	2	7	6
LLM-generated	3516	473	263	25	0	0	3779	498
Pattern expansion	257	191	40	6	0	0	297	197
Total	3862	717	316	38	74	54	4252	809

Table 2: Discourse markers breakdown. Candidates vs. found in data after corpus filtering. Manual curation and extendable connectives alone were sufficient for SPECIFY, yielding 2.8M (q, g) pairs, so we did not apply LLM generation or pattern expansion to this operation.

plates to extract hypernym–hyponym pairs, we use a template-based approach to obtain candidate markers in two stages. First, we manually construct relation-specific templates and instantiate them using curated term lists. For example, consider the common template of to [solution-term] [demonstrative] [problem-term], associated with ADDRESS. By choosing different combinations of solution and problem terms, candidates such as “to solve this limitation” and “to mitigate these risks” were instantiated. We then validate these candidates using a SOTA LLM to eliminate non-grammatical candidates. Even though this stage produced a large number of candidates (6k and 810 for ADDRESS and BROADEN respectively), most of the valid markers overlapped with the LLM-generation step. This stage yielded 397 markers, of which 297 were new; the remaining 100 had already been produced in earlier steps. We obtain 4,252 markers in total (Table 2).

Filtering stage For each sentence pair, we checked whether the second sentence begins with a marker $\ell \in L$. If so, ℓ is the prefix, the remainder is the gold sentence g , and the first sentence is the query q ; we assign the tuple (q, g) to the relation r with $\ell \in L_r$; all L_r are disjoint by construction. This yields a set of tuples per relation for each of ADDRESS, BROADEN, and SPECIFY.

We matched the full lexicon against all ~ 366.6 M consecutive sentence pairs; 809 of the 4,252 mark-

ers fired at least once, yielding $\sim 3\text{M}$ pairs across 1.1M papers. Two NLP experts independently vetted these 809 markers and agreed that every one of them was valid, qualitatively observing that all markers were clear-cut.

4.2 Temporal Split

First, we de-duplicate the dataset, ensuring that each (q, g) pair appears at most once. Pairs are then partitioned by publication date: the *training set* contains sentences from papers from 2015 through September 2025; the *validation set* the remainder of 2025. Sentences from 2026 are exclusive to the *test set* (Table 1; stage 4, Figure 2).

The temporal split was applied after de-duplicating the full dataset. It was chosen to enforce a strict approach against contamination: no test query and no test candidate could have been seen during training, and test pairs are drawn from papers published after the public release of every model we evaluate, so no backbone could have encountered them during pre-training.

The resulting benchmark comprises 3,017,476 (q, g) pairs: 2,779,177 SPECIFY, 222,707 ADDRESS and 15,592 BROADEN. The train, validation and test sets comprise three disjoint query subsets corresponding to three different ideation move retrieval tasks.

In addition, we enrich the corpus with hard negatives, introduced by markers rejected during curation (§4.1), which we name **distractor discourse markers**. These are cues that resemble valid markers but signal relations outside our three target ideation moves, as in “*On the other hand,*” (contrast), or do not explicitly refer to the immediately preceding sentence, as in “*In general,*”. The sentences they introduce are added to the candidate pool, where they are gold for no query.

In total, including the negatives, our training, validation, and test candidate pools contain 13,787,834, 361,172, and 404,371 sentences, respectively. Within each temporal partition, we use a partition-specific candidate set shared across the three retrieval tasks. This design is important for preventing a relation-specific corpus shortcut; an ADDRESS retriever, for example, cannot search an ADDRESS-only collection. Table 1 shows the corpus-size breakdown. See Appendix A for more implementation details, lexicon construction, and Appendix B.2 for temporal split.

The temporal split is designed to evaluate whether retrievers trained on earlier scientific litera-

ture can generalize to a held-out snapshot of newly published work. In preliminary non-temporal split experiments, we observed substantially higher performance, raising the possibility that models benefited from prior exposure to the evaluation texts or closely related versions. We therefore use recent papers as both test queries and candidates, reducing this risk and providing a more conservative evaluation of generalization to unseen text. The relatively short gap between training and testing also limits confounding from major shifts in the distribution of CS topics.

This protocol is not intended to reconstruct the literature available when each query’s source paper was published, but to create a conservative challenging retrieval benchmark that removes confounding factors. However, thinking more applicatively, each extracted query can be thought of as a *reusable* research problem that may be revisited at a later retrieval time; work published after the source paper may still provide valid inspiration for subsequent research—scientific problems are rarely completely solved in one paper. Our released data include publication metadata and can therefore also support alternative splits, such as cumulative literature snapshots or candidate pools restricted to work published before each query.

Although the candidate pool is not an exhaustive collection of every sentence in the source corpus, it is not restricted to sentences expressing the three target operations. Target positives constitute only about 20% of the 13.8M training candidates; the remaining approximately 80% are distractors mined using non-target discourse markers. These include contrast, similarity, continuation, inference, and other common scientific discourse patterns, spanning more than one million papers. Thus, the pool is highly heterogeneous in topic, rhetorical function, and linguistic form, and is substantially closer to broad scientific-sentence retrieval than to ranking within a small, relation-specific collection.

This construction also makes the task challenging as it acts as a form of hard-negative sampling. Candidates are coherent scientific statements extracted under the same procedure as the positives and often resemble them in topic, style, and residual discourse structure, even after the marker is removed. Moreover, because the pool is shared across operations, it contains sentences that may be topically compatible with a query but instantiate the wrong ideation move. A retriever must therefore distinguish query-conditioned relational

compatibility rather than merely reject malformed, out-of-domain, or topically unrelated sentences. By contrast, an exhaustive pool of arbitrary corpus sentences would add many easy negatives. Indeed, our preliminary experiments included candidates with many random sentences that inflated results. Nevertheless, evaluating retrieval over every sentence in the underlying corpus may be explored for testing deployment-scale coverage.

4.3 Silver Test Set Construction

Our data is mined by distant supervision. While our training set serves primarily for noisy labeled training data, to rigorously evaluate generalization we create a human-calibrated, LLM-validated *silver test set*. This is our primary evaluation set, as opposed to the “raw” test set above. We sample 10k SPECIFY test pairs by stratified sampling with equal allocation across markers, capped at each marker’s size, so that frequent markers do not dominate the sample, and take all 14,020 ADDRESS and 1,410 BROADEN test set pairs. We then instruct an LLM to go over the initial sampled set and filter out candidates. We apply strict criteria to ensure high quality (q, g) pairs. Two LLM judges first judge query and gold independently for grammar, coherence, and self-containment, i.e., interpretable without surrounding context or references to other parts of the paper. They then enforce the relation itself: for ADDRESS, we require the query to be phrased as a problem and the gold to directly mitigate that problem. BROADEN and SPECIFY require the two sentences to share a main topic rather than overlap in part of it; a BROADEN gold may sit at a higher level of abstraction, but a pair whose main themes are not clearly aligned is dropped. Direction is enforced throughout: a SPECIFY gold must be a specific case of the query rather than merely something lower in a hierarchy, and pairs are dropped when the roles are inverted or when the gold continues the topic at the same level without specifying or broadening it.

Prompt Calibration We calibrate the LLM against human annotations. In practice, we find that a single prompt may be strict on some (q, g) pairs and lenient on others; thus, for each relation we write 4–6 candidate validation prompts, and keep the two that best match expert judgments (Appendix I). We compare the prompts per relation to 300 expert annotations (100 per relation), and only keep the two highest-scoring prompts per relation. These prompts reach F_1 scores .83–.84, .86–.87, and

.76 for ADDRESS, SPECIFY, and BROADEN respectively. 60 pairs were additionally labeled by four NLP experts; inter-annotator F_1 agreement scores among the five experts were .87, .90, and .82 for ADDRESS, SPECIFY, and BROADEN respectively. Inter-annotator and prompt–annotator rates are high for an intrinsically difficult annotation task.

LLM Validation We apply SOTA LLM using the two retained prompts per relation to the initial sampled test set. A pair is kept only if both prompts accept it. The final result is a high-quality test set with a total of 17,579 queries and corresponding gold candidates (see Appendix E for more details).

5 Experiments and Evaluation

In this section, we evaluate several pre-trained and fine-tuned retrieval models on RATIO. We conduct analysis of the results, including an evaluation of Top-K retrieved inspirations.

5.1 Retrievers Training

Operation-specific models We train separate retrievers s_r for each operation: the observed operation r selects a specialized retriever s_r , and that retriever ranks the same heterogeneous candidate corpus used by all three operations. This reveals the requested relevance criterion, but not which candidate satisfies it for the particular query. For example, the ADDRESS expert must still distinguish candidates that address the query’s core problem from topically related candidates that are not relevant as solution directions to the problem.

We use separate retrievers to isolate the learnability of each operation and to represent a plausible modular system in which a user selects a specialized retrieval expert. A unified model receiving r as an instruction is a complementary setting supported by the benchmark and left for future work. Importantly, our results below show that even when the test pool is restricted to candidates from the requested operation, fine-tuning obtains substantial gains over a generic retriever.

Because the query sets are mined separately for each relation as naturally occurring instances in scientific papers, the present experiments evaluate specialization within each operation-specific query distribution and not a scenario where the same query is issued under multiple operations—this would require expensive annotation which is outside the scope of this paper.

Model	Setup	SPECIFY				ADDRESS				BROADEN			
		R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100
BM25	unigram	13.0	27.3	17.6	18.1	5.3	15.2	8.3	8.7	7.9	17.6	10.9	11.3
all-mpnet-base-v2	Baseline	12.3	26.6	16.9	17.5	6.1	16.5	9.2	9.7	11.6	24.3	15.6	16.1
	Tuned	28.5	54.3	37.0	37.8	12.8	31.7	18.7	19.4	16.3	32.5	21.8	22.4
modernbert-embed-large	Baseline	20.5	39.1	26.7	27.3	6.7	17.8	10.2	10.6	13.4	26.0	17.3	17.9
	Tuned	37.2	65.5	46.7	47.4	17.2	40.0	24.5	25.3	20.9	41.4	27.3	27.8
stella-en-1.5B-v5	Baseline	20.1	39.8	26.4	27.0	7.2	20.1	11.0	11.6	13.7	29.6	18.5	19.1
	Tuned	31.6	59.7	40.9	41.6	13.7	34.5	20.2	20.9	15.4	32.4	20.9	21.6

Table 3: **Silver** test set results (%), per operation ($n=7,327 / 9,668 / 584$ queries for SPECIFY/ADDRESS/BROADEN respectively). $R@k$ =Recall@ k , $M@k$ =MRR@ k . ModernBERT-embed-large and Stella use recommended prefixes.

Models We compare three dense retrievers: ALL-MPNET-BASE-V2 (Sentence-Transformers, 2021), MODERNBERT-EMBED-LARGE (Chaffin, 2025), instruction-following STELLA-EN-1.5B-V5 (Zhang et al., 2024) (SOTA on MIR (Garikaparathi et al., 2025)), as well as BM25. Each model is evaluated as a pre-trained baseline and after relation-specific contrastive fine-tuning with in-batch negatives. During relation-specific contrastive training, the in-batch negatives are positive candidates for other queries under the same operation. We train multiple model-prefix setups: each model with a *generic* prefix and the recommended prefix for ModernBERT-embed-large and Stella (see prefixes in Appendix C). We note that all three backbones were extensively pre-trained on scientific texts. We train each setup separately on ADDRESS, BROADEN, and SPECIFY—15 fine-tuned variations in total. BROADEN was fine-tuned on a single GPU, ADDRESS and SPECIFY on four GPUs, for a total of 1,800 GPU hours and 282 hours of evaluation and retrieval (see Appendix C for more training details).

5.2 Effect of Move-Specific Fine-Tuning

Table 3 reports results on the silver test set by ideation move (see Appendix D for the unfiltered test set, where the relative fine-tuning gains are preserved but Stella-en-1.5B-v5 leads on SPECIFY and ADDRESS). Baseline pre-trained models provide limited gains over BM25, whereas move-specific fine-tuning yields substantial improvements, increasing MRR@10 by $1.6\times$ – $2.4\times$ for ModernBERT-embed-large. ModernBERT-embed-large is the strongest model on every operation on the silver set, and its two prefix setups become on par once tuned, differing by at most 2.7 points on any operation and metric reported.

Interestingly, relative improvement does not necessarily correlate with training set size. ADDRESS

gains the most in relative terms ($\sim 2.4\times$) on 7.5% of SPECIFY’s training pairs, while BROADEN has both the smallest training set and the smallest absolute gain (+10.0). Absolute performance is low: Even the best model fails for most ADDRESS and BROADEN queries, and MRR@100 exceeds MRR@10 by at most 0.8 points, so these failures are not near-hits. These results show the task is far from solved.

Oracle operation filtering. In an ablation analysis, we filter each candidate pool to include only candidates from the specific query operation. We then evaluate the base retriever vs. the trained model only on this set. The purpose of this analysis is to rule out that trained models learn a potential shortcut: superficially detecting operation-style cues, and then ranking based on shallow topical matching. We find the trained model retains large gains over base retrievers in this setup, too (see full results and details in Appendix F.2).

5.3 Top-K Candidate Evaluation

Our silver test evaluates whether the retriever ranks the known gold sentence highly. The general problem of potential false negatives in IR in general and inspiration retrieval in particular (Garikaparathi et al., 2025) is widely known; we thus additionally evaluate by judging Top-K retrieved candidates. This allows us to determine whether fine-tuning learns to retrieve actual instances of ADDRESS/SPECIFY/BROADEN with respect to a given query, or whether the training only improves reconstruction of the discourse-mined continuation sentence. We directly judge retrieved pairs: For ADDRESS, the judge requires the candidate to directly target the query’s core problem. For SPECIFY and BROADEN, the judge explicitly rejects candidates that are merely topically related, or candidates that have the wrong directionality (e.g., more specific when the target move is broadening).

This evaluation complements the oracle operation filtering ablation mentioned above: it tests whether the resulting outputs actually satisfy the requested relation for the particular query, rather than being merely topically related or having the wrong directionality. As we detail below, fine-tuning raises performance metrics (e.g., MRR@10, Precision@10) substantially. These gains are computed without showing the judge the mined gold positive, discourse marker, candidate rank, or source model. Therefore, whatever internal decomposition the retriever uses, fine-tuning increases the frequency with which its outputs respect the requested relation.

Construction Each top-10 candidate is independently judged for whether it constitutes a valid solution (resp. broader/more specific formulation) of the query. For each ideation-move relation we sample 400 queries from the silver test set by stratified sampling with equal allocation across markers, capped at each marker’s size, 1200 in total. For each query, the judge (using the same human-calibrated prompts in §4.3) independently evaluates each (q, c) pair among the top 10 candidates retrieved by the selected fine-tuned retriever and by its pre-trained baseline. Acceptance by the judge is therefore a human-anchored decision: the prompts explicitly require the candidate to be a valid, self-contained solution that directly targets the stated problem (resp. a genuine broadening/instantiation), and were selected for agreement with a five-expert panel, matching inter-expert agreement levels (F1 .87/.90/.82 vs. prompt–annotator .73–.91; Table 10 in the Appendix). The judge is not shown the gold, the candidate’s rank, or the other candidates. We run two prompts on each (q, c) pair: *Hard* agreement requires both prompts to accept a candidate; *soft* agreement requires either. Per-prompt results appear in Table 15. We judge 24K (q, c) pairs (1,200 queries $\times$ 10 candidates $\times$ 2 systems, baseline and tuned), each under both prompts, for 48K judgments in total. In this setup a query can have multiple valid candidates other than the gold, so we report Hit Rates (fraction of queries with at least one accepted candidate in the top 10), Precision, NDCG, and MAP, all at rank 10. Full sampling, calibration, and prompt details appear in Appendix E. **Model selection** Candidate-level validation is costly and cannot cover all 15 trained models and their respective baselines. We select ModernBERT-embed-large with the recommended search_query prefix setup: it is the strongest

Relation	Model	HR@10	P@10	N@10	M@10	MAP@10
SPECIFY	Baseline	65.5	11.4	45.9	41.0	38.0
	Tuned	89.0	23.0	68.5	64.1	58.3
ADDRESS	Baseline	40.5	6.1	25.9	21.9	20.7
	Tuned	76.5	16.7	48.6	41.2	37.0
BROADEN	Baseline	19.3	2.2	13.4	11.6	11.4
	Tuned	29.0	3.6	18.7	15.9	15.1

Table 4: Top-10 retrieved candidate evaluation under hard LLM prompt agreement (%). 400 queries per relation. Model=ModernBERT-embed-large; showing HitRate@10, Precision@10, nDCG@10, MRR@10 and MAP@10. Full results in Table 14.

model on all three relations on the silver test set and substantially smaller than the 1.5B-parameter alternative, and its prefix variants are within 2.6 MRR@10 after fine-tuning on silver test set. These results characterize one retriever before and after fine-tuning.

5.4 Results and Analysis

The judge scores the top-10 candidates per query (Table 4). Under hard agreement, fine-tuned ModernBERT-embed-large (recommended prefix) reaches at least one accepted candidate for 89.0% of SPECIFY and 76.5% of ADDRESS queries, returning 1.67 accepted candidates per ADDRESS list against 0.61 for baseline.

Across all settings, the tuned ModernBERT-embed-large retrieves candidates with lower lexical overlap with the query (Jaccard over non-stopword tokens) than its baseline counterpart. Further, the tuned SPECIFY model’s top-10 candidates overlap with the query *less* than the gold does (−.019 on silver, −.029 on the judged sample); rank correlates negatively with overlap in every setting (Spearman $\rho \approx -.20$ and $-.28$). Tuned models more often retrieve pairs the judge accepts, with the largest gain on SPECIFY (Recall@10 of 63% on accepted vs. 39.1% on rejected pairs; Appendix F). We also find that for ADDRESS and BROADEN lexical overlap with gold or LLM-accepted candidates is lower than for SPECIFY, explaining its higher results (Tables 3-4).

The mined positive is not the only valid inspiration For each query we record whether the gold was retrieved in the top-10 and whether the judge accepted the top candidates. Our evaluation shows that the large majority of accepted candidates originate from papers other than the query’s, i.e., tuned retrievers surface cross-paper candidates that satisfy the requested relation under

our candidate-level judgment—“inspiration” under our operational definition.

We find that the accepted candidate is often not the mined adjacent-sentence positive. For the tuned ADDRESS retriever, 41.2% of queries have an accepted alternative in the top 10 despite the mined positive being absent: the model often finds a valid candidate other than the original gold; the corresponding rates are 24.0% for SPECIFY and 12.0% for BROADEN. These discovered positives overlap (Jaccard) with the query less than the mined positives do (e.g., .086 vs. .124 on ADDRESS; all $p < .001$), and are not near-duplicate restatements. We additionally find that 88–90% of *all* top-10 accepted candidates other than the mined gold are cross-paper for the tuned models across all three relations. Single-positive scores underestimate performance – counting any accepted candidate **raises ADDRESS MRR@10 from 24.5 to 41.2** (full analysis in Appendix G).

Thus, candidate-level evaluation shows that the model retrieves valid alternatives from the corpus, beyond the query’s paper, rather than merely reconstructing query sentence continuation. Although the model often does not retrieve the adjacent gold, it nevertheless retrieves a valid alternative, and that alternative is usually from another paper. The oracle filtering ablation and top-10 analyses provide complementary evidence. Together, our results indicate that models have learned meaningful patterns and generalizations beyond shallow shortcuts. Operation-associated style and adjacency-specific cues may therefore provide a coarse prior, but cannot account for the results. This does not imply that rhetorical compatibility is irrelevant—indeed, it is part of the supervision—but that operation style or adjacency alone cannot explain the results.

We next test the complementary question of whether candidates retrieved by trained models are also preferred under an assessment of inspiration potential.

5.5 Evaluation of Inspiration Potential

We perform a preliminary exploration of whether fine-tuning improves the *inspiration potential* of the retrieved list as a whole. Motivated by pairwise LLM-as-judge evaluation in prior inspiration-retrieval work (Garikaparthi et al., 2025), for each of the 400 queries per relation in the RATIO ranking validation set, an LLM judge is given the query sentence and two ranked lists of top retrieved candidates, one from the fine-tuned retriever and one

from the base model, labeled only as List A and List B. It is asked which list has more *inspiration potential* — which would be more useful as a source of ideas for addressing the problem raised in the query (ADDRESS), making it more specific (SPECIFY), or broadening it (BROADEN) — and answers A, B, or tie. Every query is judged twice with the list labels swapped to account for position bias. We assign a system preference only when both orders select the same underlying system. Full prompts and further details appear in Appendix H. The fine-tuned retriever is preferred for all three operations, in particular for ADDRESS (see Table 16, Appendix H). At the same time, performance remains limited, especially for BROADEN, and indicates that the task is far from solved.

6 Conclusion

We introduced RATIO, a benchmark for scientific inspiration retrieval where relevance depends on ideation roles. We establish ideation operation-conditioned retrieval and provide scalable resources for exploring it. RATIO focuses on three operations: ADDRESS retrieves responses to problems; BROADEN retrieves more general formulations; and SPECIFY retrieves more concrete instances. RATIO contains over 3 million query-positive pairs from full-text papers, using vetted discourse markers. Human-calibrated LLM judgments provide a higher-quality silver test set.

Our experiments show that generic retrievers do not capture these ideation moves, while move-specific fine-tuning substantially improves all models. Tuned models frequently retrieve valid inspirations other than the mined positive, especially for ADDRESS. At the same time, overall performance leaves substantial room for improvement.

Although we instantiate RATIO for three ideation moves in CS, our methodology can be extended to other relations and other domains, offering a general recipe for operation-conditioned retrieval supervision. Future work could extend the task beyond adjacent sentences and CS, collect extensive (costly) human annotations, and integrate these retrieval operations into end-to-end models and systems for scientific reasoning and idea development.

References

Anirudh Ajith, Mengzhou Xia, Alexis Chevalier, Tanya Goyal, Danqi Chen, and Tianyu Gao. 2024. [LitSearch: A retrieval benchmark for scientific literature](#)

- search. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15068–15083. Association for Computational Linguistics.
- Anthropic. 2026. [Claude \(4.6 opus version\)](#). Large language model.
- Lukas Biewald. 2020. [Experiment tracking with weights and biases](#). Software available from wandb.com.
- Antoine Chaffin. 2025. [ModernBERT-embed-large](#).
- Kees Dorst and Nigel Cross. 2001. [Creativity in the design process: Co-evolution of problem–solution](#). *Design Studies*, 22(5):425–437.
- Matthijs Douze, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Lucas Hosseini, Leïla Lowengrub, and Hervé Jégou. 2024. The Faiss library. *arXiv preprint arXiv:2401.08284*.
- Aniketh Garikaparathi, Manasi Patwardhan, Aditya Sanjiv Kanade, Aman Hassan, Lovekesh Vig, and Arman Cohan. 2025. [MIR: Methodology inspiration retrieval for scientific research problems](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 28614–28659, Vienna, Austria. Association for Computational Linguistics.
- Gabriela Goldschmidt. 2014. *Linkography: Unfolding the Design Process*. The MIT Press, Cambridge, MA.
- Google. 2026. [Gemini](#). Large language model.
- Chenyang Gu, Jiahao Cheng, Meicong Zhang, Pujun Zheng, Jinquan Zheng, and Guoxiu He. 2026. [MoRI: Learning motivation-grounded reasoning for scientific ideation in large language models](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 34838–34869. Association for Computational Linguistics.
- Marti A. Hearst. 1992. Automatic acquisition of hyponyms from large text corpora. In *Proceedings of the 14th conference on Computational linguistics*, volume 2, pages 539–545.
- Matthew Henderson, Rami Al-Rfou, Brian Strope, Yunhsuan Sung, László Lukács, Ruiqi Guo, Sanjiv Kumar, Balint Miklos, and Ray Kurzweil. 2017. [Efficient natural language response suggestion for smart reply](#). *Preprint*, arXiv:1705.00652.
- Tom Hope, Ronen Tamari, Hyeonsu B Kang, Daniel Hershcovich, Joel Chan, Aniket Kittur, and Dafna Shahaf. 2022. [Scaling creative inspiration with fine-grained functional aspects of ideas](#). *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*.
- Xiang Hu, Hongyu Fu, Jing Wang, Yifeng Wang, Zhikun Li, Renjun Xu, Yu Lu, Yaochu Jin, Lili Pan, and Zhenzhong Lan. 2025. [Nova: An iterative planning framework for enhancing scientific innovation with large language models](#). In *Annual Meeting of the Association for Computational Linguistics*.
- Hervé Jégou, Matthijs Douze, and Jeff Johnson. 2017. [Faiss: A library for efficient similarity search and clustering of dense vectors](#).
- Rodney Michael Kinney, Chloe Anastasiades, Russell Authur, Iz Beltagy, Jonathan Bragg, Alexandra Buraczynski, Isabel Cachola, Stefan Candra, Yoganand Chandrasekhar, Arman Cohan, Miles Crawford, Doug Downey, Jason Dunkelberger, Oren Etzioni, Rob Evans, Sergey Feldman, Joseph Gorney, David W. Graham, F.Q. Hu, and 29 others. 2023. [The semantic scholar open data platform](#). *ArXiv*, abs/2301.10140.
- Yujie Liu, Zonglin Yang, Tong Xie, Jinjie Ni, Ben Gao, Yuqiang Li, Shixiang Tang, Wanli Ouyang, Erik Cambria, and Dongzhan Zhou. 2026. [Research-Bench: Benchmarking LLMs in scientific discovery via inspiration-based task decomposition](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 13187–13207. Association for Computational Linguistics.
- Xing Han Lù. 2024. [Bm25s: Orders of magnitude faster lexical search via eager sparse scoring](#). *ArXiv*, abs/2407.03618.
- OpenAI. 2026. [ChatGPT \(version GPT-5.4\)](#). Large language model.
- Marissa Radensky, Simra Shahid, Raymond Fok, Pao Siangliulue, Tom Hope, and Daniel S Weld. 2026. Scideator: Human-llm compound system for scientific ideation through facet recombination and novelty evaluation. In *Proceedings of the ACM Conference on AI and Agent Systems*, pages 348–374.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-bert: Sentence embeddings using siamese bert-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.
- Stephen Roller, Douwe Kiela, and Maximilian Nickel. 2018. [Hearst patterns revisited: Automatic hypernym detection from large text corpora](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 358–363, Melbourne, Australia. Association for Computational Linguistics.
- Mobashir Sadat and Cornelia Caragea. 2022. [SciNLI: A corpus for natural language inference on scientific text](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7399–7409, Dublin, Ireland. Association for Computational Linguistics.

- Mobashir Sadat and Cornelia Caragea. 2024. [MSciNLI: A diverse benchmark for scientific natural language inference](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1610–1629. Association for Computational Linguistics.
- Sentence-Transformers. 2021. [all-mpnet-base-v2](#).
- Damien Sileo, Tim Van De Cruys, Camille Pradel, and Philippe Muller. 2019. Mining discourse markers for unsupervised sentence representation learning. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3477–3486.
- Noy Sternlicht and Tom Hope. 2026. [CHIMERA: A knowledge base of scientific idea recombinations for research analysis and ideation](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1871–1905, San Diego, California, United States. Association for Computational Linguistics.
- Nir Sweed, Hanit Hakim, Ben Wolfson, Hila Lifshitz, and Dafna Shahaf. 2025. Finding your MUSE: Mining unexpected solutions engine. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 30419–30434. Association for Computational Linguistics.
- Diane Tchuindjo, Devavrat Shah, and Omar Khattab. 2026. Obliq-bench: Exposing overlooked bottlenecks in modern retrievers with latent and implicit queries. *EMNLP*.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2019. [Representation learning with contrastive predictive coding](#). *Preprint*, arXiv:1807.03748.
- Rosni Vasu, Chandrayee Basu, Bhavana Dalvi Mishra, Cristina Sarasua, Peter Clark, and Abraham Bernstein. 2025. [HypER: Literature-grounded hypothesis generation and distillation with provenance](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 25413–25438. Association for Computational Linguistics.
- Qingyun Wang, Doug Downey, Heng Ji, and Tom Hope. 2024. [SciMON: Scientific inspiration machines optimized for novelty](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 279–299. Association for Computational Linguistics.
- Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Nathan Cooper, Griffin Adams, Jeremy Howard, and Iacopo Poli. 2024. [Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference](#). *Preprint*, arXiv:2412.13663.
- Dun Zhang, Jiacheng Li, Ziyang Zeng, and Fu-Yun Wang. 2024. [Jasper and stella: distillation of sota embedding models](#). *ArXiv*, abs/2412.19048.

A Relation-Marker Vocabulary: Full Construction

Note about annotators Marker vetting was done by two NLP experts: a graduate student in computer science with text-annotation experience, and a university professor. Pair-level annotation for judge calibration was done by a separate group, described in Appendix E.

Marker Lexicon Generation In an iterative process, three models — Claude Opus 4.6 (Anthropic, 2026), Gemini 3 Flash (Google, 2026), and GPT-5.4 (OpenAI, 2026) — received the task definitions, the relation subcategories, and the Hearst-inspired seeds, and generated variants tagged *strong-true*, where the marker signals the relation independently of the surrounding context; *weak-true*, where it does so only in some contexts; or *false*, where it does not signal the relation. ADDRESS subcategories included alleviate, avoid or eliminate, bypass, mitigate, and solve. Three model families were used so that the lexicon does not inherit the phrasing preferences of a single model.

Pattern expansion We expanded the discourse markers using Hearst-inspired templates. Hearst (1992) observed that certain lexico-syntactic patterns reliably signal an is-a relation in running text. For example, in “[NP₁] such as [NP₂]”, NP₁ is almost always a hypernym of NP₂. Such patterns can therefore mine hypernym–hyponym pairs from a corpus without supervision (Roller et al., 2018).

We apply the same idea to discourse relations. Each Hearst-inspired template consists of fixed text and typed slots, for example the ADDRESS template to [solT] [dem-pron] [probT]. Slots were filled from per-type term lists, and NLTK, spaCy, and PyInflect expanded each filled string into its morphological variants, so one template yields every grammatical form of its verbs and nouns. GPT-5.4 then filtered the resulting candidates (Table 5).

For ADDRESS, templates combining 116 problem words, 81 response words, and 18 modifiers generated roughly 6K candidates, of which 257 novel markers were valid, beyond manual and LLM-generated markers. For BROADEN, two generalization verbs (*generalize*, *broaden*) over ten case nouns (*observation*, *result*, *finding*, *analysis*, *case*, *phenomenon*, *insight*, *experiment*, *example*, *problem*) produced 810 candidates and 40 valid markers. SPECIFY was not expanded with tem-

Pattern	Example
<i>Address</i>	
(in order) to [response] this ([problem])	in order to mitigate this limitation
[response]-ing this ([problem])	addressing this challenge
a ([modifier]) [response]-noun for this ([problem])	a possible solution for this issue
<i>Broaden</i>	
(to) [generalize]-ing this	generalizing this
to [generalize] this [case]	to generalize these observations
[broaden cue] this	zooming out from this
[report verb] this {more generally more broadly}	suggests this more generally
this [report verb] {in general a more general}	this indicates a more general

Table 5: Examples for expansion patterns per relation. Square brackets mark slots filled from curated word lists, (parentheses) mark optional elements, {braces} mark a choice of one. *this* stands for any demonstrative (*this*, *that*, *these*, *those*), agreeing in number with the following noun. Patterns are stored lemmatized and expanded by inflection, but listed here as inflected for clarity.

plates.

Embedding discovery The pipeline extracted 9.5M sentence-initial comma-terminated phrases of at most five words. ALL-MPNET-BASE-V2 retrieved ten neighbors for each of 82 foundational markers (picked from Manual and extendable), producing 820 candidates. GPT-4 validation yielded roughly 30 additional markers, although overlap with LLM-generated and pattern-expansion-based markers limited their incremental contribution; therefore we do not report them.

Distractor Discourse Markers Distractor markers are discourse markers which were rejected from any of the three tested relations — ADDRESS, BROADEN, SPECIFY. They often yield sentences that resemble target examples in genre, location, and extraction characteristics but do not instantiate the target relation.

Most Distractor markers fall into two categories. First, overly general markers that do not reliably refer to the immediately preceding sentence (e.g. *therefore*, which can often point back to an earlier paragraph). Second, markers signaling relations outside our target set, which are very common in our corpus, and could be classified as contrast (*in contrast*) or similarity (*similarly*, *in other words*).

1,000 markers were validated as distractors by an NLP expert, and the rest by an LLM.

Published lexicon The full lexicon is publicly available and includes the three target relations and distractor subcategories for contrast, similarity, and common but irrelevant markers for public use.

B Data Extraction Details and Figures

B.1 Text normalization and cleaning

Before pairing, we apply regex-based heuristics to each sentence:

- **Normalization:** lowercasing for consistent downstream processing.
- **Noise removal:** malformed text, invalid characters, redundant whitespace, and stray list markers (bracketed numbers, leading ordinals, 1)-style prefixes).
- **Introduction-clause removal:** leading discourse markers in the antecedent sentence (e.g. *however, first, in particular*).
- **Gold-sentence prefix cleaning:** ordinal and list prefixes in gold sentences (e.g. *firstly, second*) that would distort the target, producing clean continuations.
- **Short-sentences removal:** Sentences shorter than 35 characters were dropped (all such cases were inspected manually).

Each stored tuple carries metadata including corpusid, publication date, field labels, and section category. We later ran several analyses on the discarded prefixes, none of which surfaced anything of note. A worked example appears in Figure 3.

B.2 Temporal Split: Full Rationale and Composition

To make the benchmark more challenging, pairs of (q, g) derived from distractor markers are also added to the corpus pool (See Appendix A).

Boundaries balance three constraints: an earlier training cutoff would place evaluation inside likely model pretraining windows; allocating all of 2025 to validation would overinflate it after rapid growth in computer-science publishing; and a single-year test partition starves BROADEN. Partitioning by marker rather than by date produced inflated metrics, consistent with exposure to older evaluation papers. Full details in Table 6.

C Training, Evaluation, Objectives, and Hyperparameters

Models and software The experiments use ALL-MPNET-BASE-V2², MODERNBERT-EMBED-LARGE³ (Warner et al., 2024; Chaffin, 2025), and STELLA-EN-1.5B-V5⁴ (Zhang et al., 2024) with sentence-transformers (Reimers and Gurevych, 2019).

Loss We use Multiple Negatives Ranking Loss with in-batch negatives for contrastive learning (Henderson et al., 2017; van den Oord et al., 2019):

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cos(q_i, p_i)}}{\sum_{j=1}^N e^{s \cos(q_i, p_j)}} \quad (4)$$

with the library default scale $s = 20$, i.e., $\tau = .05$.

Metric definitions For one positive per query with rank r_q ,

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{1}[r_q \leq k], \quad (5)$$

which coincides with Accuracy@ k , and

$$\text{MRR@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathbb{1}[r_q \leq k]}{r_q}. \quad (6)$$

Setups We trained and evaluated a total of 15 combinations across relations and setups. All three models with a **generic prefix** ("query: " for query/candidate sentences and "document: " per gold); ModernBERT-embed-large with additional **search query prefix** (search_query:/search_document:); and Stella with additional system-recommended instruction (Instruct: Given a web search query, retrieve relevant passages that answer the query. Query:). Appending the marker at the end of the query was evaluated on smaller subsets, yielded slightly better but similar results to the generic setup, so ultimately rejected as unrealistic; per-relation instructions gave no gain.

Baselines BM25 used unigrams, unigrams+bigrams, and unigrams+trigrams, with English stopword removal and Snowball stemming,

²<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

³<https://huggingface.co/lightnait/modernbert-embed-large>

⁴https://huggingface.co/NovaSearch/stella_en_1.5B_v5

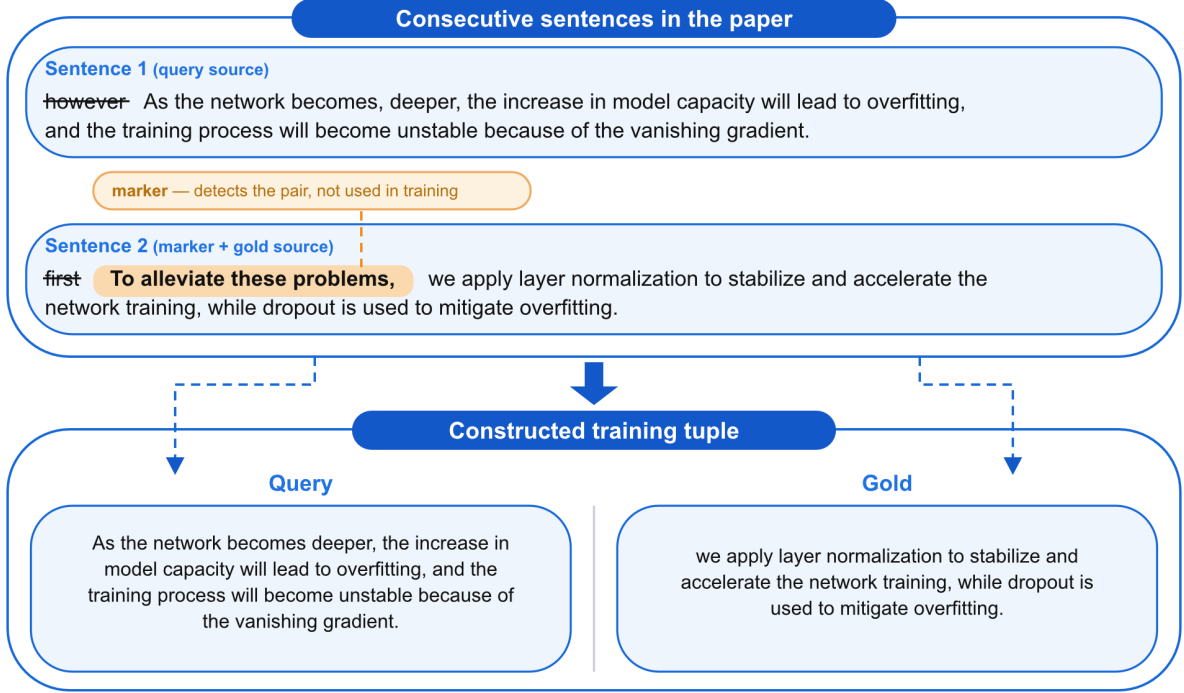


Figure 3: Worked extraction example, including marker removal and cleaning of the query–marker–positive triple.

Partition	Train			Validation			Test		
	Queries	Gold	Candidates	Queries	Gold	Candidates	Queries	Gold	Candidates
<i>Target relations</i>									
SPECIFY	2,605,515	2,605,515		79,583	79,583		94,079	94,079	
ADDRESS	195,605	195,605		13,082	13,082		14,020	14,020	
BROADEN	13,687	13,687		495	495		1,410	1,410	
<i>Distractors</i>									
Explicit contrast	–	1,135,357	13,787,834	–	48,963	361,172	–	63,086	404,371
Implicit similarity	–	15,440		–	426		–	401	
Explicit similarity	–	567,983		–	12,636		–	12,518	
Common irrelevant	–	9,254,247		–	205,987		–	218,857	
All	2,814,807	13,787,834	13,787,834	93,160	361,172	361,172	109,509	404,371	404,371

Table 6: Full Temporal split. Each query is paired with a single gold; distractors contribute golds to the candidate corpus but are never issued as queries. The candidate corpus of each split is shared by all relations and equals its total gold count.

implemented with BM25S (Lù, 2024) and a Faiss index (Jégou et al., 2017; Douze et al., 2024). Precision@ k equals Recall@ k/k , and nDCG is a monotone discount of the same rank; MRR@10 is therefore primary. In Table 8 M@100 is reported to show that performance is not merely a matter of rank depth. It exceeds M@10 by at most 0.8 points for every system.

Optimization Each one of the 15 combinations used a manual search of three to five hyperparameter configurations, seeded from published recommendations (See Table 7). Checkpoints were evaluated after each epoch using validation

MRR@10. The best validation checkpoint was restored. The best trial per setup was likewise selected on validation MRR@10. The test set was used once, for final reporting. SPECIFY and ADDRESS used four NVIDIA L40S GPUs with cross-device negative gathering (DDP). Runs were tracked with Weights & Biases (Biewald, 2020).

Training results These results show the task is far from solved. With ModernBERT-embed-large using the recommended prefix, MRR@10 reaches 39.1 on SPECIFY, 21.5 on ADDRESS and 22.2 on BROADEN (Table 8), with the mined positive reaching the top ten for 56.7%, 35.4% and

34.6% of queries respectively. $\text{MRR}@100$ exceeds $\text{MRR}@10$ by at most 0.8 points, so the misses do not sit just below the cutoff.

Interestingly, the input prefix matters most for ModernBERT-embed-large in baseline, where the recommended prefix gains 3.3 $\text{MRR}@10$ points on ADDRESS over the generic one; the gap nearly vanishes after tuning (0.1). For Stella-en-1.5B-v5 the two prefixes are indistinguishable in baseline (0.0) and differ by 0.6 once tuned.

LLM as a judge We use GPT-5.4 as an automatic judge via the OpenAI Batch API, which halves per-token cost relative to synchronous calls. Each request places the full judging prompt (Appendix I) in the system message and the instance to score — (q, g) or (q, ℓ, g) — in the user message, with responses constrained to JSON (`response_format=json_object`), `reasoning_effort=medium`, and `verbosity=low`.

D Full Automatic Results

Full retrieval automatic results on the **test** set, per relation and setup, are reported in Table 8, silver test set in Table 9.

E Relevance-Judge Calibration and Construction

Note about annotators Pair-level annotation was done by five graduate students in computer science, all with research and text-annotation experience. Annotators received the same prompts as the LLM judge.

Calibration One hundred pairs per relation were manually annotated for coherence, direction, and validity by an NLP expert. Four to six prompts per relation were tested and the two highest- F_1 prompts were retained (Table 10).

Retrieved candidate set construction Four hundred silver queries per relation were sampled using the same allocation procedure. For SPECIFY, the sample came from an earlier confidence-filtered subset of 5,707 pairs, although all remain valid under the final silver rule. The retriever’s top 100 candidates were retained and its top 10 judged. The judge did not observe the mined positive, rank, or other candidates. Silver input was `querySentence`. `discourse-marker`, `goldSentence`; ranking validation set input was

Model	#	Ep	Batch	LR	Reg. / Rationale
BROADEN					
mpnet	0	4	64	2e-5	– / best-so-far
	1	3	64	2e-5	wd .01 / published
	2	5	64	3e-5	– / higher capacity
	3	4	16	1e-5	wd .01, frz .15 / anti-overfit
	4	4	32	1e-5	– / best-practice
ModernBERT	0	3	32	2e-5	– / best-so-far
	1	1	32	2e-5	wd .01, wu .06 / Nomic recipe
	2	5	32	3e-5	– / higher capacity
	3	4	16	1e-5	wd .01, frz .15 / anti-overfit
	4	3	32	8e-5	mgn .5 / probe
stella	0	5	16	2e-5	mgn .5 / higher capacity
	1	5	16	5e-6	frz .75 / anti-overfit
	2	2	16	1e-5	mgn .5 / best-so-far
	3	2	16	2e-6	– / reference
	4	1	4	5e-5	wu .05 / reference
ADDRESS					
mpnet	0	1	32	3e-5	– / prior winner
	1	2	32	3e-5	– / prior winner, more epochs
	2	1	16	2e-5	wd .01 / published
	3	1	32	1e-5	– / probe
ModernBERT	0	1	24	3e-5	– / prior winner
	1	2	24	3e-5	– / prior winner, more epochs
	2	1	16	8e-5	mgn .5 / probe
	3	1	16	2e-5	wd .01, wu .06 / Nomic recipe
stella	0	1	16	2e-6	– / prior winner
	1	2	16	2e-6	– / prior winner, more epochs
	2	1	16	2e-5	mgn .5 / improve
	3	1	4	5e-5	wu .05 / reference
SPECIFY					
mpnet	0	1	64	2e-5	– / prior regime
	1	1	16	2e-5	wd .01 / published
	2	1	32	3e-5	– / improve
ModernBERT	0	1	24	2e-5	– / prior regime
	1	1	24	3e-5	– / improve
	2	1	16	2e-5	wd .01, wu .06 / Nomic recipe
stella	0	1	16	5e-6	wd .01, wu .05, mgn .5 / GTE recipe
	1	1	8	2e-5	wd .01, wu .05, mgn .5 / improve
	2	1	4	5e-5	wu .05 / reference

Table 7: Hyperparameter configurations per ideation operation. Negatives = batch size $\times$ world size (1 for BROADEN and four for ADDRESS and SPECIFY). wd=weight decay, wu=warmup, mgn=max-gradient norm, frz=freeze ratio.

`querySentence`, `candidateSentence`. Hard agreement uses conjunction; soft agreement uses disjunction.

F Extended Analyses

F.1 Lexical Overlap

For each query, mean word-level Jaccard overlap was computed between the query and top- k candidates and compared with query-positive overlap. Text was lowercased, tokenized on word characters, stripped of English stopwords, and Snowball-stemmed, matching the BM25 representation.

Tuned retrieval has lower overlap than off-the-shelf retrieval in all 36 comparisons over $k \in \{1, 3, 5, 10\}$, three operations, and three query sets. On SPECIFY, tuned top-10 overlap falls below the positive reference: $-.010$ on test, $-.019$ on sil-

System Setup		SPECIFY				ADDRESS				BROADEN			
		R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100
Lexical baselines													
BM25-unigram	—	10.6	23.3	14.6	15.1	5.0	14.0	7.7	8.1	6.5	14.2	8.8	9.2
BM25-bigram	—	8.4	19.8	11.9	12.4	4.4	12.7	6.9	7.3	5.7	13.6	8.0	8.3
BM25-trigram	—	7.0	18.7	10.5	11.0	3.7	11.9	6.1	6.5	5.0	13.1	7.3	7.6
all-mpnet-base-v2	generic-Baseline	10.0	22.7	13.9	14.4	5.7	15.5	8.6	9.1	9.8	20.4	13.1	13.6
	generic-Tuned	25.2	48.9	32.8	33.6	11.3	28.3	16.6	17.3	13.4	26.8	17.7	18.2
modernbert-embed-large	generic-Baseline	15.1	32.1	20.4	21.0	4.0	11.9	6.4	6.8	11.0	23.5	14.9	15.4
	generic-Tuned	30.5	56.5	39.0	39.8	14.8	35.5	21.4	22.2	15.8	34.4	21.7	22.3
	recommended-Baseline	16.3	33.0	21.7	22.2	6.5	16.8	9.7	10.1	11.8	23.3	15.5	16.0
	recommended-Tuned	30.5	56.7	39.1	39.8	15.0	35.4	21.5	22.2	16.4	34.6	22.2	22.7
stella-en-1.5B-v5	generic-Baseline	16.3	34.3	22.0	22.6	6.8	19.0	10.5	11.0	11.7	26.0	16.0	16.6
	generic-Tuned	31.9	59.0	40.8	41.5	15.7	37.6	22.7	23.5	15.6	32.1	20.7	21.5
	recommended-Baseline	16.2	34.2	21.9	22.5	6.8	18.9	10.5	11.0	11.7	25.8	16.0	16.5
	recommended-Tuned	32.2	59.2	41.1	41.8	16.3	38.4	23.3	24.1	14.8	32.2	20.1	20.8

Table 8: Retrieval automatic results on the **test** set (%), per relation. $R@k$ =Recall@ k ; $M@k$ =MRR@ k ; $M@10$ is the primary metric. Each model is shown per input setup - baseline vs. trained and prefix type (recommended prefix vs. generic). All systems retrieve against the relation-shared corpus.

System Setup		SPECIFY				ADDRESS				BROADEN			
		R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100
Lexical baselines													
BM25-unigram	—	13.0	27.3	17.6	18.1	5.3	15.2	8.3	8.7	7.9	17.6	10.9	11.3
BM25-bigram	—	10.6	23.5	14.6	15.1	4.7	13.4	7.4	7.8	7.9	16.4	10.4	10.7
BM25-trigram	—	9.0	22.5	13.0	13.5	4.0	12.5	6.5	6.9	6.5	15.9	9.3	9.6
all-mpnet-base-v2	generic-Baseline	12.3	26.6	16.9	17.5	6.1	16.5	9.2	9.7	11.6	24.3	15.6	16.1
	generic-Tuned	28.5	54.3	37.0	37.8	12.8	31.7	18.7	19.4	16.3	32.5	21.8	22.4
modernbert-embed-large	generic-Baseline	18.5	38.2	24.9	25.5	3.9	11.8	6.3	6.7	13.2	27.4	17.6	18.2
	generic-Tuned	34.9	62.8	44.1	44.8	16.9	40.2	24.4	25.2	19.3	40.6	26.2	26.9
	recommended-Baseline	20.5	39.1	26.7	27.3	6.7	17.8	10.2	10.6	13.4	26.0	17.3	17.9
	recommended-Tuned	37.2	65.5	46.7	47.4	17.2	40.0	24.5	25.3	20.9	41.4	27.3	27.8
stella-en-1.5B-v5	generic-Baseline	20.0	39.9	26.5	27.1	7.2	20.2	11.1	11.6	13.5	29.6	18.4	19.0
	generic-Tuned	30.9	57.4	39.7	40.4	11.4	30.3	17.3	18.0	15.1	32.4	21.0	21.6
	recommended-Baseline	20.1	39.8	26.4	27.0	7.2	20.1	11.0	11.6	13.7	29.6	18.5	19.1
	recommended-Tuned	31.6	59.7	40.9	41.6	13.7	34.5	20.2	20.9	15.4	32.4	20.9	21.6

Table 9: Full Retrieval results on the **silver** test set (%), per relation group. $R@k$ =Recall@ k , $M@k$ =MRR@ k ; $M@10$ is the primary metric. ADDRESS column reflects the silver set (9,668 queries). Each model is presented per input setup, prefix (recommended or generic). All systems retrieve against the relation-shared candidate pool.

Relation	Source	Annotator panel			Single annotator				
		Agr.	F_1	α	Agr.	F_1	κ	P	R
ADDRESS	Human annotators	83.3	.87	.61	—	—	—	—	—
	Minimal-v2	75.0	.76	.41	79.0	.84	.55	.83	.84
	Ordered-v2	71.7	.73	.40	78.0	.83	.52	.82	.84
SPECIFY	Human annotators	86.7	.90	.67	—	—	—	—	—
	Anchor-full-v2	86.7	.91	.66	81.0	.87	.51	.82	.93
	Anchor-compact	80.0	.87	.20	79.0	.86	.46	.82	.90
BROADEN	Human annotators	83.3	.82	.68	—	—	—	—	—
	Directional	80.0	.81	.62	73.0	.76	.46	.83	.71
	Relaxed	70.0	.75	.40	70.0	.76	.36	.74	.79

Table 10: Prompt agreement with human labels. Left: 60 pairs from a five-annotator panel, averaged over annotator pairs (human row) or prompt-annotator pairs (prompt rows), so the rows are directly comparable. Right: 300 pairs from the single annotator the prompts were calibrated on. P, R and κ require a designated reference and so appear only on the right; the same reason means no human-human agreement exists there, hence the dashes. α is Krippendorff’s α . Agreement is high for this difficult annotation task.

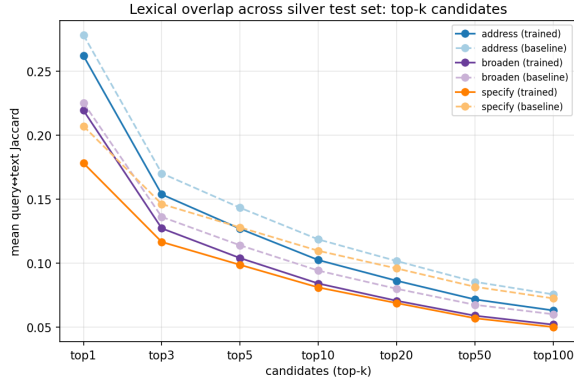


Figure 4: Query-candidate lexical overlap by relation, over retrieval depth for silver set and ModernBERT-embed-large. Lexical overlap is Jaccard score with unigram, stopwords removed, snowball-stemmed.

ver, and $-.029$ on gold. Rank-overlap Spearman correlation is negative in every relation and setup, ranging from $-.20$ to $-.28$ (Figure 4).

Relative query-gold Jaccard change from test set to silver test set can be found in Table 11.

	Content words	All words
SPECIFY	+10.0	+7.3
BROADEN	+11.6	+6.0
ADDRESS	+7.1	+4.9

Table 11: Relative query-gold Jaccard change from test to silver (%).

F.2 Oracle Operation Filtering

We test whether trained models are learning a shortcut in the form of identifying the style of operation sentences combined with generic topical similarity. For each operation we restrict the candidate pool to candidates of that operation, and re-evaluate the off-the-shelf and the fine-tuned model on the same silver queries (Table 12). We use two pools, the raw test split and the silver test set.

Operation filtering does not explain the gains. On the silver set, fine-tuning still adds 26.6, 20.1 and 15.8 MRR@10 points for SPECIFY, ADDRESS and BROADEN—larger absolute gaps than on the shared pool (20.0, 14.3 and 10.0). The relative gain on ADDRESS does fall from $2.4\times$ to $1.7\times$ under filtering, but mainly because the baseline rises with the smaller pool; pools of different size are not comparable, and each panel of Table 12 should be read within itself.

F.3 Retriever-Judge Agreement

On sampled test pairs before filtering, Recall@10 is higher for accepted than rejected pairs in every relation and model state. Fine-tuning increases the separation, particularly for ADDRESS (Table 13).

G Discovered Positives: Ranking validation set

Discovered positives overlap the query less than mined positives: (discovered vs. mined): $.075$ vs. $.137$ on SPECIFY, $.086$ vs. $.124$ on ADDRESS, and $.093$ vs. $.129$ on BROADEN. Mann-Whitney rank-biserial effects are $.46$, $.29$, and $.27$; all $p < .001$.

Favoring anaphoric relations Explicit anaphors (*this*, *these*, *such*, *it*, *the above*) appear in 93.4% of sampled ADDRESS pairs but only three SPECIFY and two BROADEN pairs. Within ADDRESS, explicit-anaphor pairs are accepted at a rate of 71.9% versus 27.7% for bare markers (Fisher OR 6.7, $p < .001$). This property is relation-specific.

H TOP-10 Judge Inspiration

Inspiration across relations. All three rates exceed chance under two-sided binomial tests. This ordering matches the *relative* MRR@10 gain of fine-tuning for ModernBERT-embed-large—approximately $2.4\times$, $1.7\times$, and $1.6\times$ for ADDRESS, SPECIFY, and BROADEN, respectively—despite the judge never seeing the gold. BROADEN is where the judge is least reliable: its verdicts flip between list orders most often (82.2% stable), and it is the only relation with a significant position effect, favoring the second-listed option (Slot A 45.4%). Table 16 breaks the inspiration judge down by relation. Prompts can be found in Appendix I.4.

I Prompts

I.1 Discourse Markers Classifier: ADDRESS, BROADEN

SPECIFY lexicon generated the largest pool with minimal discourse markers, so all were evaluated by two NLP experts, and there was no need for an LLM classifier.

You are an expert linguistic judge evaluating connective phrases in scientific text. Your job is to decide, with high certainty, whether a given connective phrase reliably signals that the sentence following it offers a SOLUTION, IMPROVEMENT, or MITIGATION that is DIRECTLY inspired by and targets the problem described in the sentence preceding it.

Candidate pool	Setup	SPECIFY				ADDRESS				BROADEN			
		R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100	R@1	R@10	M@10	M@100
(a) Relation’s own test split	Baseline	29.5	48.0	35.4	36.0	23.1	41.7	28.7	29.4	39.7	63.5	46.9	47.8
	Tuned	48.8	74.7	57.5	58.1	40.3	65.4	48.1	48.9	54.8	77.6	61.7	62.4
(b) Relation’s silver golds only	Baseline	46.0	64.9	52.0	52.6	25.4	44.7	31.1	31.8	44.7	69.7	52.6	53.5
	Tuned	72.0	90.8	78.6	79.0	43.4	68.4	51.2	52.0	61.3	84.6	68.4	69.0

Table 12: **Oracle operation filtering.** Retrieval results on the **silver** test set (%), per relation, when the candidate pool is restricted to candidates of the queried relation. All rows are ModernBERT-embed-large with the recommended (search_query) prefix, off-the-shelf (*Baseline*) and fine-tuned on the matching relation (*Tuned*). $R@k$ =Recall@ k , $M@k$ =MRR@ k ; M@10 is the primary metric. Both panels use the same 7,327, 9,668 and 584 silver queries and differ only in the pool: (a) the relation’s own test split, of 94,079, 14,020 and 1,410 candidates; (b) the relation’s silver-annotated golds only, of 7,327, 9,668 and 584 candidates, for SPECIFY, ADDRESS and BROADEN respectively. Bold marks the better of Baseline/Tuned within each panel.

Relation	Model	Accepted	Rejected	Δ	OR
SPECIFY	base	38.6	25.7	12.9	1.82
	trained	63.0	39.1	23.9	2.65
ADDRESS	base	17.9	14.5	3.3	1.28
	trained	40.0	25.3	14.8	1.97
BROADEN	base	26.0	21.4	4.6	1.29
	trained	41.4	29.8	11.7	1.67

Table 13: Recall@10 conditioned on relevance-judge acceptance top-10 (%). ModernBERT-embed-large, search_query prefix, baseline and fine-tuned.

A "problem-solution" relationship means sentence 1 describes a problem, limitation, challenge, or gap, and sentence 2 proposes an approach, method, or remedy that is directly motivated by that specific problem. This is the "to address this, we..." relationship - the solution must be a direct response to the prior problem, not merely a related contribution that happens to follow. Return True ONLY if, across the vast majority of plausible scientific contexts, the phrase forces a reading in which the following sentence is a remedy DIRECTLY inspired by the prior problem. Return False if the phrase could plausibly introduce other relationships such as generalization, specification, consequence, paraphrase, emphasis, exemplification, contrast without a tied remedy, or continuation of the problem description. When uncertain, return False. The bar for True is not "a solution follows" but "a solution clearly tied to the specific problem just stated follows." Edge cases and guidance: - The phrase must anaphorically or purposively tie the solution to the prior problem. Generic cues that do not reference the problem are False. - Contrast phrases ("however", "on the other hand", "in contrast", "nevertheless", "yet", "nonetheless") are always False - they mark opposition without committing to a remedy tied to the prior problem. - Causal/inferential phrases ("therefore", "thus") are False - they can mark consequence or takeaway, not reliably a targeted remedy. - Additive phrases ("furthermore", "moreover", "additionally", "in addition") are False - they extend discourse without tying what follows to the prior problem. - Generalization ("more generally", "more broadly"), exemplification ("for example", "for instance", "e.g."), restatement ("in other words", "that is", "i.e.", "namely"), and emphasis ("notably", "importantly", "crucially", "indeed") phrases are all False. - Bare purposive phrases without an anaphoric tie ("to address", "to overcome", "to mitigate", "to solve",	"to tackle", "to alleviate", "to circumvent", "to remedy") are False - they can head infinitival clauses for any purpose. - Bare first-person contribution verbs without an anaphoric tie ("we propose", "we introduce", "we present", "we develop", "our approach", "our method") are False - they often open contributions cold, without direct motivation from the prior sentence. - Anaphoric purposive phrases that explicitly reference the prior problem are True: "to address this", "to address this issue", "to address this problem", "to address this limitation", "to address these challenges", "to address the above", "to overcome this", "to overcome this limitation", "to overcome these limitations", "to mitigate this", "to tackle this issue", "to alleviate this", "to resolve this", "to solve this problem", "to circumvent this", "to remedy this". - Explicit solution-pivots with back-reference are True: "as a solution to this", "our solution to this", "to fix this". - Contribution verbs combined with an anaphoric tie are True (e.g., "to address this, we propose"). Output strictly valid JSON with this schema: <pre>{ "label": <true false>, "rationale": "<one sentence explaining the decision>", "confidence": <"high" "medium" "low"> }</pre> Do not include any text outside the JSON object. Do not wrap the JSON in markdown fences.
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Listing 1: Discourse Markers Classifier: Address

You are an expert linguistic judge evaluating connective phrases in scientific text. Your job is to decide, with high certainty, whether a given connective phrase reliably signals that the sentence following it is a GENERALIZATION of the sentence preceding it. A "generalization" relationship means sentence 2 broadens, abstracts, or extends the scope of sentence 1 - moving from a specific claim to a wider one. This is the "more generally" relationship. Return True ONLY if, across the vast majority of plausible scientific contexts, the phrase forces a generalization reading. Return False if the phrase is ambiguous and could plausibly introduce other relationships such as: - Specification or narrowing (opposite direction) - Implication, consequence, or takeaway - Paraphrase or restatement - Emphasis or highlighting - Example or illustration - Contrast or qualification When uncertain, return False. We prefer to discard ambiguous connectives rather than admit noisy ones into the dataset.	
---	--

Relation	Setup	Hard agreement					Soft agreement				
		HR	P	nDCG	MRR	MAP	HR	P	nDCG	MRR	MAP
SPECIFY	Recommended prefix-Baseline	65.5	11.4	45.9	41.0	38.0	78.3	16.2	54.5	49.0	44.3
	Recommended prefix-Tuned	89.0	23.0	68.5	64.1	58.3	93.3	30.1	73.1	70.6	61.9
ADDRESS	Recommended prefix-Baseline	40.5	6.1	25.9	21.9	20.7	52.8	8.5	32.4	26.8	25.1
	Recommended prefix-Tuned	76.5	16.7	48.6	41.2	37.0	86.3	20.8	55.1	47.3	41.6
BROADEN	Recommended prefix-Baseline	19.3	2.2	13.4	11.6	11.4	43.5	6.0	28.7	24.3	23.4
	Recommended prefix-Tuned	29.0	3.6	18.7	15.9	15.1	62.0	9.9	38.8	33.0	30.0

Table 14: Ranking validation results rank 10 (%) under hard agreement (both judge prompts accept) and soft agreement (either accepts), 400 queries per relation. ModernBERT-embed-large, search_query setup; -*Baseline* off-the-shelf, -*Tuned* fine-tuned on the matching relation. HR=Hit-Rate@10, P=Precision@10, nDCG=nDCG@10, MRR=MRR@10. Better of Baseline/Tuned per relation in **bold**.

Relation	Setup	Prompt 1					Prompt 2				
		HR	P	nDCG	MRR	MAP	HR	P	nDCG	MRR	MAP
SPECIFY	Recommended prefix-Baseline	71.0	13.0	50.1	45.1	41.3	73.5	14.6	50.7	45.1	41.1
	Recommended prefix-Tuned	89.8	24.6	70.0	66.4	59.7	93.0	28.5	71.8	68.5	60.6
ADDRESS	Recommended prefix-Baseline	44.2	6.8	28.0	23.6	22.2	50.2	7.8	30.8	25.3	23.9
	Recommended prefix-Tuned	80.2	18.4	50.9	43.5	38.4	83.5	19.1	53.1	45.2	40.4
BROADEN	Recommended prefix-Baseline	27.3	3.4	18.2	15.6	15.2	37.0	4.8	24.3	20.7	19.8
	Recommended prefix-Tuned	41.5	5.6	26.0	21.9	20.6	54.0	7.9	33.3	28.0	25.8

Table 15: Top-10 candidate evaluation by individual judge prompt rank 10 (%) , 400 queries per relation. ModernBERT-embed-large, search_query setup. Prompts 1 and 2 are the two human-calibrated prompts retained per relation (Table 10); Table 14 reports their conjunction and disjunction. HR=Hit-Rate@10, P=Precision@10, nDCG=nDCG@10, MRR=MRR@10.

Edge cases and guidance:

- Phrases like "in general" or "generally" are ambiguous because they can introduce a default/baseline claim rather than a genuine broadening. Return False for these unless the generalization reading clearly dominates.
- Phrases that combine generalization with another function (e.g., "more generally speaking, therefore") should be judged on the dominant function. If generalization is dominant and reliable, return True.
- Purely additive or continuative phrases ("furthermore", "moreover", "additionally") are False - they extend discourse but do not necessarily broaden scope.
- Causal or inferential phrases ("therefore", "thus", "hence") are False - they mark inference, not generalization.
- Exemplification phrases ("for example", "for instance", "such as", "e.g.") are False - they move from general to specific, the opposite direction.
- Restatement phrases ("in other words", "that is", "i.e.", "namely") are False - they paraphrase at the same level of abstraction.
- Contrast phrases ("however", "on the other hand", "in contrast", "nevertheless") are False - they mark opposition, not generalization.
- Emphasis phrases ("notably", "importantly", "crucially", "indeed") are False - they highlight salience, not broaden scope.
- Phrases explicitly marking scope expansion ("more generally", "more broadly", "in a broader sense", "in a more general sense", "at a higher level of abstraction") are True.

Output strictly valid JSON with this schema:

```
{
  "label": <true | false>,
  "rationale": "<one sentence explaining the decision>",
  "confidence": <"high" | "medium" | "low">
}
```

```
}
```

Do not include any text outside the JSON object. Do not wrap the JSON in markdown fences.

Listing 2: Discourse Markers Classifier: Broaden.

I.2 Silver Test Set: Query-Gold ADDRESS, SPECIFY and BROADEN

Input is "S1 (the problem). connecting phrase, S2 (the solution/mitigation)." Judge S1 and S2 independently, then the match.

Note: The connecting phrase is consistent with a *problem-solution* relation, but its presence does not guarantee that this is the actual relation between the sentences. Judge the relation from the sentence content itself, not from the connecting phrase alone. Ignore the connecting phrase when assessing the match.

is_problem_valid = True only if S1 states a clear, specific problem AND the core problem is articulated in S1 itself (a pronoun pointing outside S1 is OK only if it is NOT the core problem). False if garbled, if it only says a problem exists without saying what, or if the core problem hides behind an external reference. False if S1 isn't actually a problem - it's a capability, requirement, or assumption.

is_solution_valid = True only if S2 describes a specific solution/mitigation an expert would recognize from its terms, with the core solution articulated in S2 (a reference back to the problem sentence is fine). False

	Preferred		Win	Stable	Slot A
Relation	-T	-B	(%)	(%)	(%)
<i>Judge: gpt-5.4</i>					
ADDRESS	240	102	70.2	85.8	48.6
SPECIFY	217	135	61.6	88.5	50.0
BROADEN	185	144	56.2	82.2	45.4*
<i>Judge: gpt-5.6</i>					
ADDRESS	246	106	69.9	88.0	47.5
SPECIFY	214	131	62.0	86.2	50.5
BROADEN	163	148	52.4	77.8	40.6*

Table 16: Inspiration judge, 400 queries per relation (1200 total) judged in both list orders, by two judge models. *Win* is the share of stable, decisive queries preferring the fine-tuned retriever; *Stable* the share given the same verdict under both orders; *Slot A* how often the first-listed option was chosen, where 50% indicates no position bias. Significance is by two-sided exact binomial test against 0.5 ($*p < 0.01$). Win rates reach $p < 10^{-4}$ except BROADEN: $p = 0.03$ for gpt-5.4 and not significant under gpt-5.6 ($p = 0.43$).

if garbled, or if it only names/asserts a solution exists or defers details ("the following ..."). False if S2 only explains how a method works, or is a diagnostic/experimental setup, or is a consequence/derivation, not a remedy.

match = True only if both are valid AND S2 directly targets the core problem in S1.

OUTPUT (JSON only):

```
{
  "is_problem_valid": {"is_valid": "True|False", "reason": "<1 sentence>"},
  "is_solution_valid": {"is_valid": "True|False", "reason": "<1 sentence>"},
  "match_evaluation": {"is_match": "True|False", "reason": "<1 sentence>"}
}
```

Listing 3: Silver: Query sentence-Gold sentence: Address - Prompt 1.

Given a passage from scientific literature of the form "S1 (the problem). connecting phrase, S2 (the solution/mitigation).", evaluate whether it forms a valid problem-solution pair for an information retrieval test set. S1 is the problem, S2 is the solution/mitigation. Execute the steps in order.

Note: The connecting phrase is consistent with a *problem-solution* relation, but its presence does not guarantee that this is the actual relation between the sentences. Judge the relation from the sentence content itself, not from the connecting phrase alone. Ignore the connecting phrase when assessing the match.

- Problem (S1)

Return "True" only if:

 - S1 states a clear, specific problem (expert terminology an expert recognizes as a problem, OR a descriptive articulation of one). Naming a problem category without saying what makes it a problem is NOT clear and specific.
 - Self-contained: the core problem is articulated in S1. A pronoun/reference pointing outside S1 is acceptable only if it is NOT the core problem.

Return "False" if: grammatically incorrect/incoherent; it states a problem exists but not what it is; the core problem is hidden behind a reference pointing outside S1; or S1 isn't actually a problem - it's a capability, requirement, or assumption.
- Solution (S2)

Return "True" only if:

- S2 describes a solution/mitigation an expert would understand from its terms (established domain terminology counts as sufficiently described).
- Self-contained: the core solution is articulated in S2. A reference to the problem sentence is acceptable; a reference elsewhere is acceptable only if it is NOT the core solution.

Return "False" if: grammatically incorrect/incoherent; it only states a solution exists / gives a bare name / defers details ("the following ...") without saying what it is or does; or S2 only explains how a method works, or is a diagnostic/experimental setup, or is a consequence/derivation, not a remedy.

3. Match

Return "True" only if both S1 and S2 are valid AND S2 directly targets the core problem in S1. Otherwise "False".

OUTPUT (JSON only):

```
{
  "is_problem_valid": {"is_valid": "True|False", "reason": "<1 sentence>"},
  "is_solution_valid": {"is_valid": "True|False", "reason": "<1 sentence>"},
  "match_evaluation": {"is_match": "True|False", "reason": "<1 sentence>"}
}
```

Listing 4: Silver Query sentence-Gold sentence: Address - Prompt 2.

You judge whether SENTENCE_2 is "more specific" than SENTENCE_1.

Both sentences are drawn from the same scientific paper.

INPUT FORMAT

The input is "S1. <connecting phrase>, S2". The connecting phrase was chosen because it signals this relation, but it only marks the AUTHOR'S INTENT - it does not guarantee the content fits. Judge the actual sentences, not the phrase. A "specifically" / "for example" cue is a hint, not proof.

DEFINITION

S2 is "more specific" when it instantiates, elaborates, or narrows a claim made generally in S1. S2 takes the general statement and supplies a concrete case, a named example, exact figures, an enumerated set of members, or the detailed mechanism that realizes it. The direction runs from general (S1) to particular (S2).

THE CORE TEST (apply in order)

- Identify the general claim in S1.
- Ask: is S2 an INSTANCE OF, or a FINER-GRAINED ACCOUNT OF, that exact claim? It must narrow the SAME proposition, not introduce a different one. Valid forms of "more specific":
 - exemplification: S2 gives a named example of S1's general class (e.g. S1 "several studies explored X"; S2 "Smith et al. did X via Y")
 - enumeration: S2 lists the concrete members S1 generalized over (e.g. S1 "we evaluate distinct LLM families"; S2 "GPT, LLaMA, Claude...")
 - quantification: S2 gives the exact numbers behind S1's qualitative claim (e.g. S1 "model improves"; S2 "+2.4% on benchmark Z")
 - mechanism/elaboration: S2 spells out HOW S1's stated operation works (e.g. S1 "tunable embeddings replace [v]"; S2 "'a photo of a [v] dog' becomes 'a photo of a [v_l] dog' per layer")
 - worked case/scenario: S2 grounds S1's principle in one concrete situation.

ANCHOR CHECK

S2's referring expressions ("this", "it", "these results", "the method") must resolve cleanly to S1's content. If S2's anchor instead points to the paper as a whole ("this work shows...") and is NOT recoverable from S1, the pair is loose - decide false.

DECIDE true (more specific) ONLY IF ALL HOLD:

- S1 states something more general than S2.
- S2 narrows the SAME proposition (instance / detail / enumeration / numbers / mechanism), not a different or sibling proposition.
- Both sentences are individually coherent and not too vague to assess.

- S2's anchor resolves to S1.

DECIDE false IF ANY HOLD:

- S2 is MORE GENERAL than S1 (abstracts it) - that is the inverse relation.
- S1 and S2 are PARALLEL/SIBLING points at the same level: two coordinate effects, two advantages, two future-work directions, a contrast ("however", "in contrast"), or two restatements of one claim.
- ELABORATION/JUSTIFICATION: S2 explains the mechanism or objective rather than giving a narrower case of S1, or S2 justifies why S1 holds rather than narrowing it.
- CO-LEVEL RESULT: S1 is a result/performance claim and S2 is co-level evidence or numbers (same-level result reporting), rather than a general claim that S2 instantiates.
- S2 introduces NEW content (a different mechanism, a contrasting concept, a separate finding) rather than narrowing S1.
- S2 is an INFERENCE/CONSEQUENCE drawn from S1 rather than a concrete instance of it.
- The two sentences are on DIFFERENT TOPICS with no instance relation.
- Either sentence is garbled, truncated mid-clause, or too vague to evaluate - if so, decide false and say so in the reason.

PARTIAL SPECIFICATION

If S1 makes a multi-part claim and S2 instantiates only ONE part, this still counts as true, but note the partial coverage in the reason and lower confidence slightly.

MIXED S2

If S2 begins by specifying S1 but then adds a new/contrasting clause, label on the DOMINANT move; if specification leads, true with a note; if the new material dominates, false.

OUTPUT (JSON only, no preamble, no markdown):

```
{"decision": true|false, "confidence": <float 0.0-1.0>,
  "reason": "<one short sentence>"}
```

Listing 5: Silver Query sentence-Gold sentence: Specify - Prompt 1.

You judge whether SENTENCE_2 is "more specific" than SENTENCE_1.

INPUT FORMAT

The input is "S1. <connecting phrase>, S2". The connecting phrase was chosen because it signals this relation, but it only marks the AUTHOR'S

INTENT - it does not guarantee the content fits. Judge the sentences, not the phrase. A "specifically" / "for example" cue is a hint, not proof.

DEFINITION

S2 is "more specific" when it narrows the SAME claim S1 makes generally - via a concrete example, an enumeration of the members S1 generalized over, exact figures behind a qualitative claim, or the detailed mechanism that realizes it. Direction: general (S1) -> particular (S2).

TEST: is S2 an instance of, or a finer-grained account of, S1's exact claim?

Decide true only if: S1 is more general; S2 narrows the SAME proposition; both sentences are coherent and not too vague; S2's references resolve to S1.

Decide false if: S2 is more GENERAL than S1 (inverse); the two are PARALLEL/sibling points (two effects, two advantages, a contrast, a restatement); S2 adds NEW content or is an inference/consequence rather than an instance; the topics differ; or either sentence is garbled or too vague.

If S1 is multi-part and S2 instantiates only one part, still true - note the partial coverage and lower confidence.

If S2 specifies then adds new material, label on the dominant move.

OUTPUT (JSON only, no preamble or markdown):

```
{"decision": true|false, "confidence": <float 0.0-1.0>,
  "reason": "<one short sentence>"}
```

Listing 6: Silver Query sentence-Gold sentence: Specify - Prompt 2.

You are evaluating sentence pairs from scientific literature for an information retrieval benchmark.

You are given a single string of the form "S1. connecting term, S2". Determine whether S2 is a *more-general* expansion of S1.

Note: The connecting term is consistent with a *more-general* relation, but its presence does not guarantee that this is the actual relation between the sentences. Judge the relation from the sentence content itself, not from the connecting term alone. Ignore the connecting term when assessing direction and scope.

Definition of the "more-general" relation:

S2 expands on S1 by stating something broader, higher-level, or more general than S1. S1 is the specific; S2 broadens it. Broadening is meant inclusively - S2 counts as more general if it does ANY of the following relative to S1:

- abstracts S1 into a wider principle, category, or claim;
- states the cross-domain applicability of S1 ("not specific to X; any domain where ... can do the same");
- frames S1's specific result/method at the paper level - its broader significance, contribution, or implication;
- draws a broader implication, recommendation, or future direction from S1;
- states a related but higher-level claim that S1's content falls under.

The relation is DIRECTIONAL: S1 is the specific, S2 is the broader/higher-level statement. It does NOT require S1 to be a literal instance of S2's exact proposition - a looser "S1 is an aspect of / contributes to / motivates the broader statement S2" is acceptable, as long as S2 is genuinely broader and connected to S1.

Evaluate the criteria below in order. Return 'true' only if ALL hold:

1. Grammar/coherence: Both S1 and S2 are grammatically correct and coherent on their own.
2. Connection: S2 directly continues from or refers back to S1. If S2 contains references ("this", "these", "such", "our findings", "the principles demonstrated here", "table N", etc.), they must resolve to content present in S1 - not to a prior sentence, a figure/table, or external context.
3. Direction & scope: S2 is broader / higher-level than S1 in any of the inclusive senses above.

Return 'false' if ANY of the following hold (checked in the same order):

- GRAMMAR: Either sentence is grammatically incorrect, garbled, or incoherent.
- CONNECTION: S2 refers to a sentence other than S1, or to figures/tables/findings not present in S1, or its references cannot be resolved from S1 alone.
- INVERSE: S2 is MORE specific than S1 (the inverse relation). A pair may be coherent and well-connected and still fail here.
- SAME LEVEL: S2 merely restates S1 or continues the topic at the same level of specificity without broadening scope.
- UNRELATED: S2 is not topically connected to S1.

Do NOT reject solely because S2 frames things at the paper level ("this work / the approach / the benchmark shows ..."), broadens to other domains, or draws a broader implication - these are valid forms of more-general here, provided S2 is broader than S1 and its references resolve to S1.

Reason field:

- If 'false' because either sentence is grammatically incorrect or incoherent, set "reason" to exactly "grammar_fail".
- Otherwise, give one short sentence naming the failing criterion: inverse / same-level / unrelated / connection.

Output format: Return only a JSON object:

```
{
  "decision": true/false, "confidence": [0-1], "reason":
    "<one short sentence>"
}

Input:
{input_string}
```

Listing 7: Silver Query sentence-Gold sentence: Broaden - Prompt 1.

You are evaluating sentence pairs from scientific literature for an information retrieval benchmark. You are given a single string of the form "S1. connecting term, S2". Determine whether S2 is a **more-general** expansion of S1.

Note: The connecting term is consistent with a **more-general** relation, but its presence does not guarantee that this is the actual relation between the sentences. Judge the relation from the sentence content itself, not from the connecting term alone. Ignore the connecting term when assessing direction and scope.

Definition of the "more-general" relation:
S2 expands on S1 by stating something broader, higher-level, abstracting, or more general than S1. That is, S1 expresses a specific case, instance, or detail, and S2 generalizes from it to a wider principle, category, or claim. The relation is DIRECTIONAL: S1 is the specific, S2 is the general.

Evaluate the criteria below in order. Return 'true' only if ALL hold:

- Grammar/coherence: Both S1 and S2 are grammatically correct and coherent on their own.
- Connection: S2 directly continues from or refers back to S1. If S2 contains references ("this", "these", "such", "the above", etc.), they must resolve to content present in S1 - not to a prior sentence or external context.
- Direction & scope: S2 is strictly broader in scope than S1. S1 must read as the specific case and S2 as the generalization of it.

Return 'false' if ANY of the following hold (checked in the same order):

- GRAMMAR: Either sentence is grammatically incorrect, garbled, or incoherent.
- CONNECTION: S2 refers to a sentence other than S1, or its references cannot be resolved from S1 alone (assumes missing context).
- INVERSE: S2 is MORE specific than S1 (this is the inverse relation, not more-general). A pair may be coherent and well-connected and still fail here.
- SAME LEVEL: S2 merely continues the topic at the same level of specificity without broadening scope.
- UNRELATED: S2 is not topically connected to S1.
- Other: Any other reason that prevents S2 from being a more-general expansion of S1.

Note: criteria 2 (connection) and 3 (direction) are independent. A pair can be correctly connected via anaphora yet still be same-level or inverse - such pairs are 'false'.

Reason field:

- If the decision is 'false' because either sentence is grammatically incorrect or incoherent, set "reason" to exactly "grammar_fail".
- Otherwise, give one short sentence explaining the decision (e.g., naming the failing criterion: inverse / same-level / unrelated / connection).

Output format: Return only a JSON object:
{
 "decision": true/false, "confidence": [0-1], "reason":
 "<one short sentence>"
}

Input:
{input_string}

Listing 8: Silver Query sentence-Gold sentence: Broaden - Prompt 2.

I.3 Top 10 Candidate Evaluation Prompts

Prompts for top-10 candidates evaluation for ADDRESS, SPECIFY and BROADEN are identical to silver prompts (two per relation), with only two changes across all prompts and relations:

- "NOTE" about connecting term (discourse marker) is dropped.
- The sentence "*You are given a single string of the form 'S1. connecting term, S2'.*" is replaced with "*You are given s1 – a query sentence, and s2 – a candidate sentence.*"

I.4 Inspiration Judge: ADDRESS, SPECIFY and BROADEN

Unlike the silver and top-10 prompts, the input format is not described in the prompt itself: the system message holds only the task, and the user message carries only the data, in the form shown in Listing 9. Lists are labelled A and B only, so the judge is blind to which retriever produced which.

Query: <query sentence>

Candidate List A:
1. <candidate sentence>
...
5. <candidate sentence>

Candidate List B:
1. <candidate sentence>
...
5. <candidate sentence>

Listing 9: Inspiration Judge: user message format.

You are helping a researcher who is looking for their next research move.

The researcher wrote a query. You will be shown that query and two ranked lists of retrieved candidates from scientific papers, List A and List B.

Decide which list has more INSPIRATION POTENTIAL for the researcher: which list would be more useful to them as a source of ideas for addressing the problem, limitation, or challenge raised in the query.

Judge each list as a whole.

Reply with JSON only, in exactly this shape:
{
 "winner": "<A or B or tie>", "confidence": <number from 0 to 1>, "reason": "<one short sentence>"
}

- "winner" must be one of: "A", "B", "tie".
- "confidence" is how sure you are of that choice, from 0 to 1.
- Use "tie" only when the two lists are genuinely equivalent in inspiration potential.

Listing 10: Inspiration Judge: Address.

The SPECIFY and BROADEN prompts are identical to Listing 10, with one change across both: the phrase "*addressing the problem, limitation, or challenge raised in the query*" is replaced with

- SPECIFY: *“making the query more specific or concrete”*
- BROADEN: *“broadening or generalizing the query”*